



Software Release Notes for the Acuitas OS Image, Release 3.0.5/3.0.6

Acuitas Release 9.4

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Contents

- Audience.....3**
- How to Obtain Product Documentation3**
- Technical Assistance Center3**
- Purpose of the Document4**
- Software in This Release.....4**
 - Versioning Scheme4
 - Checking File Integrity4
- Requirement to Upgrade the OS5**
- Known Anomalies5**
 - New Minor Anomalies.....5
- Resolved Issues6**
- OS Image Security Updates6**
- Ports Used by the Acuitas System7**
 - Incoming Ports Opened for Acuitas Server Firewall Configuration7
 - Ports Opened Between Workstation and Acuitas Server.....7
 - Ports Opened Between Acuitas Server and Network Element8
 - Ports Opened Between Acuitas Server and Other Servers8
 - Ports Requiring Connectivity Between Acuitas Servers (HA).....9

Release Abstract

This document contains the software release notes for the Affirmed Networks distribution of the Acuitas OS software.

Audience

This document is intended for network operators, installation staff, and other qualified service personnel responsible for installing and updating the Acuitas OS software.

How to Obtain Product Documentation

Documentation is available for currently supported product releases. Documentation is available online in Adobe PDF format. You can view PDFs online using the Adobe Reader[®]. To download the latest version of the Adobe Reader software from the Adobe web site, go to <http://get.adobe.com/reader/>

Product documentation is available from the Affirmed Networks SFTP server. Contact your sales representative or the TAC for instructions on accessing the SFTP server.

See the *Affirmed Networks Documentation Library* for a complete list of Affirmed Networks product documentation.

Technical Assistance Center

Affirmed Networks Technical Assistance Center (TAC) provides technical support 24 hours a day, 7 days a week (24 x 7) for any customer with an active maintenance contract.

To contact the TAC:



E-mail: prodsupport@affirmednetworks.com



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Purpose of the Document

This Acuitas release supports Red Hat Enterprise Linux 7.7 and MySQL 8.0.

This document provides a summary of Acuitas software delivery files and versions, open issues, resolved issues, resolved vulnerabilities, and OS upgrade considerations.

The following software is included in the image:

- Kernel Version (3.10.0-1062.1.2.el7.x86_64)
- MySQL Version (8.0.18-1.1)
- Java Version (OpenJDK Runtime Environment (build 1.8.0_222-b10))

Software in This Release

The following files are included in this release:

- an-ems-3.0.5-rhel-7.7-mysql-8.0.18.iso
- an-ems-3.0.6-rhel-7.7-mysql-8.0.18.ova
- an-ems-3.0.5-rhel-7.7-mysql-8.0.18.qcow2

Versioning Scheme

The image filename contains the three versions of software types in the release:

- Affirmed Networks upgrade package version, for example, 3.0.0.
- MySQL Database version, for example, 8.0.18.
- Red Hat Enterprise Linux version. For example, 7.7.

Checking File Integrity

Use the **sha1sum** Linux command on the ISO file and verify the output matches the values in the sha1 file.

```
unix# sha1sum an-ems-3.0.5-rhel-7.7-mysql-8.0.18.iso
d04d84bad85f980222f13413eb6b0ef012556ef3
an-ems-3.0.5-rhel-7.7-mysql-8.0.18.iso
```

```
unix# sha1sum an-ems-3.0.6-rhel-7.7-mysql-8.0.18.ova
fd1ec3e4716574d442faa42806bed3d150a0ed65
an-ems-3.0.6-rhel-7.7-mysql-8.0.18.ova
```

```
unix# sha1sum an-ems-3.0.5-rhel-7.7-mysql-8.0.18.qcow2
```

```
60a7a2d1c110d942a10a9a7c16437e8885e7adcf
an-ems-3.0.5-rhel-7.7-mysql-8.0.18.qcow2
unix# cat an-v2.0-osp13-rhel-server-7.6-x86_64.iso.sha1
394b52b02eb9921e4100db1736ba7526bd03c2cb
an-v2.0-osp13-rhel-server-7.6-x86_64.iso
```

Requirement to Upgrade the OS

If you are upgrading Acuitas from a release prior to 9.4.0.0 to release 9.4.0.0 or later, that upgrade requires an upgrade to the OS. For instructions on how to perform the OS upgrade, see *OS Upgrade Considerations* in the *Acuitas System Administration Guide*.

Known Anomalies

The following known anomalies exist in this release.

New Minor Anomalies

The following minor anomalies exist in this release.

Minor Anomaly	Description
47645 Installing an-ems-3.0.6-rhel-7.7-mysql-8.0.18.ova may display as product version 3.0.5 in VMWare	The ems-3.0.6-rhel-7.7-mysql-8.0.18.ova is correctly installed. You can ignore the benign display.

Resolved Issues

This section lists the resolved issues in this release.

Red Hat ID	Description
40681 Using vm_setup script to resize the VM disk size causes a benign error	Using <code>ems_setup.sh</code> (formerly <code>vm_setup.sh</code>) script to resize the VM disk size no longer returns a benign I/O error.
41046 The vm_setup.sh option to skip host disk resizing does not start mysqld, causing Acuitas installation to fail	The <code>ems_setup.sh</code> (formerly <code>vm_setup.sh</code>) option to skip the host disk resizing now correctly starts <code>mysqld</code> .
REQS-8327, REQS-8366, REQS-9132, 45562, 45563, 46791 Security vulnerabilities	Removed several security vulnerabilities
REQS-8894 Acuitas upgrade to RedHat OpenJdk 8.0	Acuitas uses RedHat OpenJdk 8.0 (java-1.8.0-openjdk-1.8.0.222.b10-1.el7_7.x86_64) for EMS Java runtime
REQS-9169, 45912 Requirement to reduce the number of audit log files	Acuitas reduced the number of kept audit log files from 100 to 50. This change reduces the maximum steady-state disk usage by audit logging below the existing threshold.

OS Image Security Updates

Nessus security scan reports are available upon request. This release includes the security updates included in RHEL 7.7.

Ports Used by the Acuitas System

This section lists the ports opened by the Acuitas system.

Incoming Ports Opened for Acuitas Server Firewall Configuration

Table 1 Incoming Ports Opened for Acuitas Server Firewall Configuration

TCP Port	UDP Port	Used by
22		OpenSSH
80	80	Apache httpd. Traffic is redirected to port 443.
	123	NTPD
	161	SNMP daemon
162	162	SNMP trap receiver
443		Secure (HTTPS) connection
3306		MySQL server
8989	8989	Remote jconsole (Java Monitoring and Management Console)

Ports Opened Between Workstation and Acuitas Server

Table 2 Ports Opened Between Workstation and Acuitas Server

Workstation	Acuitas	Used by
	22	OpenSSH
	80	Apache httpd
	443	Secure (HTTPS) connection
	3306	MySQL server
	8989	Remote jconsole (Java Monitoring and Management Console)

Ports Opened Between Acuitas Server and Network Element

Table 3 Ports Opened Between Acuitas Server and Workstation

Acuitas	Network Element	Used by
22	22	OpenSSH
161	161	SNMP daemon
162	162	SNMP trap receiver
	830	Netconf (MME)
	2022	Netconf (MCC)
	2222	SFTP (MCC)

Ports Opened Between Acuitas Server and Other Servers

Table 4 Ports Opened Between Acuitas Server and Other Servers

Acuitas	Other Servers	Used by
	25	Mail Server for unauthenticated SMTP
123	123	NTP Server
1161	161	OSS that uses SNMP Inform
	162	OSS that receives Acuitas traps
	1514	Syslogd that received syslog streamed from Acuitas

Ports Requiring Connectivity Between Acuitas Servers (HA)

Table 5 Ports Requiring Connectivity Between Acuitas Servers (HA)

Port	Used by
1098	Java RMI
1099	RMI server socket
2224	Pacemaker
3121	Pacemaker
3306	MySQL server
3528	Corba IIOP Service
3873	Jboss EJB3 Remoting
4444	Jboss RMI/JRMP invoker
4445	Jboss Pooled Invoker
4446	Jboss Remote invoker
4712	Microsoft Distributed Transaction Coordinator
4713	JbossTS Transaction Status Manager
5405	Pacemaker
5445	Jboss hornetq
7600	Jboss Cluster Peer Discovery
8009	Apache Jserv
8083	JBoss service httpd
9990	Remote jconsole (Java Monitoring and Management Console)
9999	Pacemaker

Jboss and Pacemaker open additional ports for cluster communication.